# A Variable Data-Window Phaselet Transformer Differential Protection Algorithm Based on Recursive Least Square Theory

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*Abstract*—**With the application of high and extra-high voltage transformer, traditional differential protection algorithm based on half period data window will possess a switching interval of 30ms or even longer, which could hardly satisfy the quickness of protective relaying. When external fault occurs, inceptive fault transient component, inrush current, CT saturation, and harmonics would amplify the unbalance current, which easily bring about the malfunction of differential protection. This paper proposes an algorithm which based on recursive least square theory, and combine with variable data window phaselet algorithm to improve transformer differential protection. This least square based method is adaptive, and modulate the length of data window dynamically, and is accompanied by the continuous outlet frequency judgment, in order to ensure the correctness in the interferences of inrush current and inner fault. Simulation shows that algorithm is able to boost the reliability and sensitivity of transformer differential protection, and is significantly practicable than traditional methods.**

***Keywords*：*Transformer; Differential Protection; Least Square; Phaselet; Filter;Variable Data Window***

## I. INTRODUCTION

Current differential protection for transformer works simply, sensitively, and quickly, and has been widely applied as the main scenario in transformer protection. Nevertheless, in practical operation, inrush currents, CT saturation, and system operation mode all contribute to the malfunction of protection, and lead to a, at most, 75% operation precision rate, which is far from satisfaction[1]. To overcome the demerits listed above, the method which based on the combination of half and whole cycle data window algorithm is applied. Consequently, in the circumstance of interior faults, the interval of protection operation would last 30ms; in external faults or fault clearing, thanks to aperiodic components and CT transience, the unbalance current would be amplified and lead to a easily malfunction of protection. Currently, a myriad of novel algorithms have been proposed domestically and abroad, adaptive second harmonic restrained method[2],transient data using mathematical morphology[3], morpgological filtering[4], equivalent circuit equilibrium equation[5], error estimation[6], Time-Domain Algorithm of Capacitive Current Compensating[7] , symmetrical waveform[8],to name just a few.

Relay protection, based on acquisition of fault components timely and accurately, requires a even shorter data window. Recently, GE in America has proposed a novel digital algorithm--the phaselet algorithm, which uses sample value as the partial sum of Fourier algorithm, ignores the restriction of integer times of semi-period data-window length to recover amplitudes and angels of any integer harmonics[9]. This paper bases on the quickness of phaselet algorithm, uses recursive least square and error indicator adjust the length of data-window to improve the preciseness and quickness of protection algorithm. Moreover, the resistibility of CT saturation when external fault CT saturates and inrush current occurs, the algorithm avoid the malfunction effectively.

## II. TRADITIONAL TRANSFORMER RATIO RESTRAINT OF DIFFERENTIAL PROTECTION

Transformer phase-separated ratio restraint of differential protection often uses current phasor. Considering the direction that current flows into transformer as positive direction, ratio restraint of differential protection could be represented by equation followed:

$$I_d = |\dot{I}_1' + \dot{I}_2'| \quad I_z = |\dot{I}_1' - \dot{I}_2'|/2 \tag{1}$$

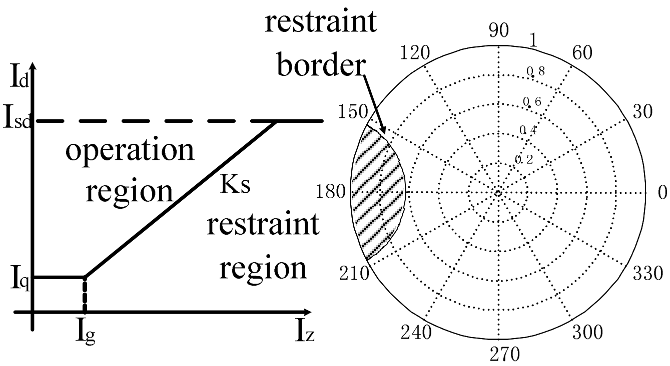
where $I_d$ represents the action current, $I_z$ represents the restraint current

the operation criterion is:

no restraint region: $I_d \geq I_q$ when $I_z \leq I_g$ (2)

ratio restraint region: $I_d \geq K_s(I_z - I_g) + I_q$ when $I_z \geq I_g$ (3)

action region: $I_d \geq I_{sd}$ when $I_z \leq I_g$ (4)




(a)ratio restraint character (b) circle restraint character

Figure 1. ratio restraint character and circle restraint character

Figure 1(a) consists from three sections, no restraint region, ratio restraint region, and action region. If current amplitude

This work was the Project 50977012 supported by National Natural Science Foundation of China (NSFC).

is lower than $I_g$ ,the action current is constant, which is $I_q$ in the figure; If current amplitude is larger than $I_g$ , the action current increases along with a bias; If current amplitude is larger than $I_{sd}$ , the fault is considerably significant, protection would act immediately. Figure 1(b) shows the action character represented by a circle. The peak current value $\dot{I}_m$ is the base value, and $\dot{\rho} = \dot{I}_n / \dot{I}_m$ , consequently, $|\dot{\rho}| \leq 1$ is the normalized value. In any circumstances, the value of relative amplitude-angle relationship should not beyond the edge of the circle. Thus, analysis of action character transformed into the analysis of action edge inside the circle. The shadow part in Figure 1(b) represents the restraint region.

## III. PHASELET ALGORITHM AND WHOLE CYCLE FOURIER ALGORITHM

In extra high voltage system, with the inductance character of transmission line and electromagnetic character of transformer, components in fault current are basically high frequency harmonics, fundamental harmonic, and offset aperiodic components, and offset aperiodic components are often filtered by front LP filter.

Least square method means to fit the input transient data with a given model, which consists from aperiodic components and certain harmonics, using least square theory. Consequently, certain amplitude and angel parameters wanted would be gained. The fitting model could be set as follow:

$$x(n) = \sum_{k=1}^{K} X_k \sin(k\omega_1 t + \theta_k) + X_0 e^{-\lambda t} \tag{5}$$

where $X_k$ is the amplitude of $k$ th harmonic, $X_1$ is amplitude of fundamental component, $\omega_1$ is angular frequency of fundamental components, $\theta_k$ is angel of $k$ th harminic, $X_0$ is the initial value of offset DC, $\lambda$ is decaying time constant. To linearized (5)

$$x(n) = X_0 - X_0\lambda t + \sum_{k=1}^{K} X_k \sin(k\omega_1 t)\cos\theta_k + \sum_{k=1}^{K} X_k \cos(k\omega_1 t)\sin\theta_k \tag{6}$$

Suppose there are $N$ sample points in one cycle, assuming the sampling points to obtain the value of $N$ ,sampling values are $x(1)$ , $x(2)$ , …… $x(N)$ , the matrix could be made as follow:

$$AY = X \tag{7}$$

Where:

$$A = \begin{bmatrix} 1 & t_1 & \sin\omega_1 t_1 & \cos\omega_1 t_1 & \cdots & \sin K\omega_1 t_1 & \cos K\omega_1 t_1 \\ 1 & t_2 & \sin\omega_1 t_2 & \cos\omega_1 t_2 & \cdots & \sin K\omega_1 t_2 & \cos K\omega_1 t_2 \\ \vdots & \vdots & \vdots & \vdots & \cdots & \vdots & \vdots \\ 1 & t_N & \sin\omega_1 t_N & \cos\omega_1 t_N & \cdots & \sin K\omega_1 t_N & \cos K\omega_1 t_N \end{bmatrix}$$

$$Y = \begin{bmatrix} X_0 \\ -X_0\lambda \\ X_1\cos\theta_1 \\ X_1\sin\theta_1 \\ \vdots \\ X_K\cos\theta_1 \\ X_K\sin\theta \end{bmatrix} \qquad X = \begin{bmatrix} x(1) \\ x(2) \\ x(3) \\ x(4) \\ \vdots \\ \vdots \\ x(N) \end{bmatrix}$$

the number of the unknown is 12, therefore 12 sample points are required, in other words, $N$ >=12. Suppose $N$ =12, A would be a constant matrix, and $Y$ could be transformed as follow:

$$Y = A^{-1}X \tag{8}$$

Often, $N$ >12, thus, data window should be extended to increase the preciseness.

$$Y = A^{+}X \tag{9}$$

where $A^{+}$ is the pinv Moore-penrose of $A$ ,

$$A^{+} = (A^T A)^{-1} A^T \text{。} \tag{10}$$

Whole cycle Fourier algorithm is the special case of least square. If the model set in least square only include fundamental components, fitting function could be rewritten as follow:

$$x(n) = \sum_{k=1}^{K} X_k \sin(k\omega_1 t + \theta_k)$$

(11)

Consequently, $A$ of $AY = X$ is:

$$A = \begin{bmatrix} 0 & 1 \\ \sin\frac{2\pi}{N} & \cos\frac{2\pi}{N} \\ \vdots & \vdots \\ \sin t_N \frac{2\pi}{N} & \cos t_N \frac{2\pi}{N} \end{bmatrix}$$

(12)

Phaselet algorithm was proposed by GE, and is actually applied, it ignores the restriction of integer times of semi-cycle. Phaselet algorithm is another case of least square. Separate each cycle into $N$ phaselets, where $N$ is an even, and each phaselet has $i$ sample points, thus the number of phaselets in one cycle is $N/i$ , the $p$ th phaselet could be represented by the following matrix:

Suppose

$$A = \begin{bmatrix} 0 & 1 \\ \sin\frac{2\pi}{N} & \cos\frac{2\pi}{N} \\ \vdots & \vdots \\ \sin(i-1)\frac{2\pi}{N} & \cos(i-1)\frac{2\pi}{N} \end{bmatrix} \tag{13}$$

$$(A^T A)^{-1} = \begin{bmatrix} \sum_{(p-1)i+1}^{pi} \sin^2 \frac{2\pi n}{N} & \sum_{(p-1)i+1}^{pi} \sin\frac{2\pi n}{N}\cos\frac{2\pi n}{N} \\ \sum_{(p-1)i+1}^{pi} \sin\frac{2\pi n}{N}\cos\frac{2\pi n}{N} & \sum_{(p-1)i+1}^{pi} \cos^2 \frac{2\pi n}{N} \end{bmatrix}^{-1}$$

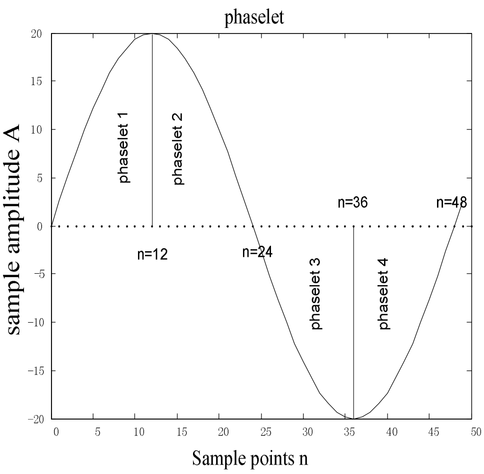
(14)

make $(A^T A)^{-1} = \begin{bmatrix} B_{RR} & B_{RI} \\ B_{RI} & B_{II} \end{bmatrix}^{-1}$ ,put it into (14),then

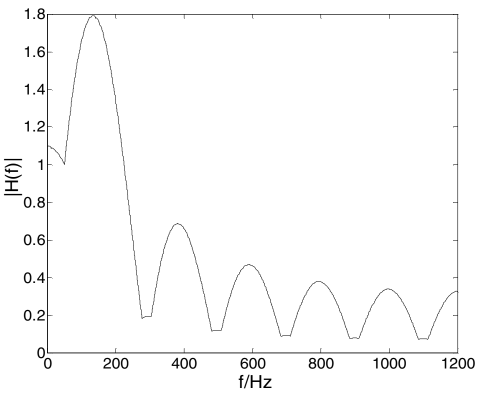
$$\hat{Y} = \begin{bmatrix} B_{RR} \sum_{(p-1)i+1}^{pi} x(n)\cos\frac{2\pi n}{N} + B_{RI} \sum_{(p-1)i+1}^{pi} x(n)\sin\frac{2\pi n}{N} \\ B_{RI} \sum_{(p-1)i+1}^{pi} x(n)\cos\frac{2\pi n}{N} + B_{II} \sum_{(p-1)i+1}^{pi} x(n)\sin\frac{2\pi n}{N} \end{bmatrix}$$

(15)

where $B_{RR}$ 、 $B_{RI}$ 、 $B_{II}$ are the filtering coefficients of pahselet, $x(n)$ is the sampled signal. Pheselet diagram is shown in Figure 2(a),character of pheselet is shown in Figure 2(b).




(a)pheselet diagram




(b)pheselet filter character

Figure 2. pheselet diagram and filter character

From the analysis above, the conclusion could be like this: the process of phaselet algorithm is weighted summing[10-12]. Every single sample in each phaselet multiplied with a given matrix $(A^T A)^{-1}$ creates a phaselet, on the contrary, each phaselet multiplied with $(C^T C)^{-1}$ it would transform back to the traditional phasor.

## IV. VARIABLE DATA-WINDOW PHASELET ALGORITHM

### 4.1 Variable Data-Window Phaselet Algorithm Based on Recursive Least Square Theory

The use of phaselet algorithm can significantly speed transformer operation, however the filtering performance is poor due to the short data window. Although this error can be avoided by increasing the number of outlet judgment and adapting the self-adaptive value of protection,appropriately eliminating the impact of error of short data window algorithm for filtering, but the probability of protection malfunction is still relatively high. Adaptive variable data window can be used in this case. After the failure occurred, filter data window length will self-adaptively increase along with the increasing of sampling points, and the sampling precision will be improved as the data window length is increasing. In microcomputer protection, the sampled data is based on the sampling frequency provided to the computer one by one, the sampling data keep increasing along with the growing length of the data window, the phaselet algorithm must re-calculate once per new sample data added, causing much calculation. If the original estimated value is amended on a new sample data's being taken, this will constitute a recursive least squares algorithm.

Make $X(n+1)$ as the new sampling data of self-adaptively increased data window, and the improved parameter estimation vector $Y(n+1)$, then

$$Y(n+1) = \left[H^T(n+1)H(n+1)\right]^{-1} H^T(n+1)X(n+1) \qquad (16)$$

where

$$H(n+1) = \begin{bmatrix} H(n) \\ h(n+1) \end{bmatrix}, \quad X(n+1) = \begin{bmatrix} X(n) \\ \mathrm{x}(n+1) \end{bmatrix}$$

suppose

$$P(n) = \left[H^T(n)H(n)\right]^{-1} ,$$

$$P(n+1) = \left[H^T(n+1)H(n+1)\right]^{-1}$$

$$= P(n) - P(n)h^T(n+1)\left[h(n+1)H(n)h^T(n+1)+1\right]^{-1} h(n+1)P(n) \qquad (17)$$

put equation (17) into equation (16), and a recursive least squares estimation in the form of square is obtained.

$$Y(n+1) = Y(n) + K(n+1)\left[X(n+1) - h(n+1)Y(n)\right] \qquad (18)$$

where

$$K(n+1) = P(n)h^T(n+1)\left[h(n+1)P(n)h^T(n+1)+1\right]^{-1}$$

It can be seen from equation (18) that the new estimate vector can be estimated from original vector added by a correction term, and in fact this term reflects the forecasting error brought about when the sampling value of current sampling time is forecasted by the original estimated vector.

Traditional CT has two saturation problems: steady-state saturation and transient saturation. CT will not immediately enter the saturation on the early stages of failure, because the increase in flux is the integral of the current of time. At this time CT is in the linear region, but when the fault current increase to a certain extent, CT will be enter saturation because of the rapidly increasing secondary excitation current. When CT steady-state saturation occurs, the secondary current and excitation current are non-sine wave which contains a lot of high harmonics, there are steady-state errors of mass change in CT. More cases of failure, CT is prone to enter the transient saturation because of the decaying DC components. In this case, CT's time of entering and quitting saturation are different each cycle. CT will gradually withdraw from saturated along with the decaying of DC components of each current and the reverse back induction of power frequency within each period, and once again secondary current is able to accurately reflect the primary current. The primary current and secondary current are shown in Figure 3:

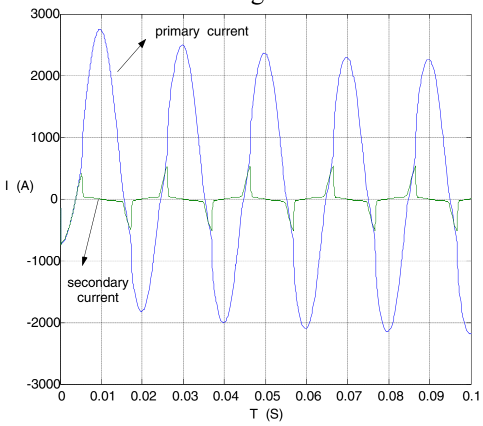
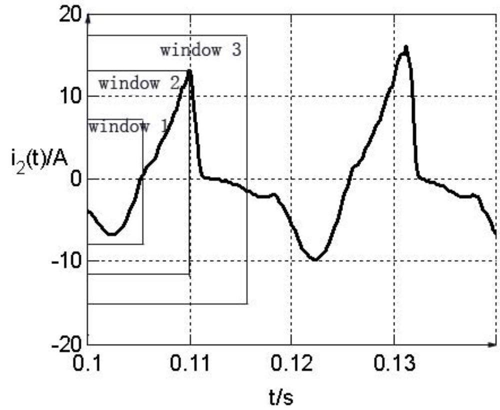
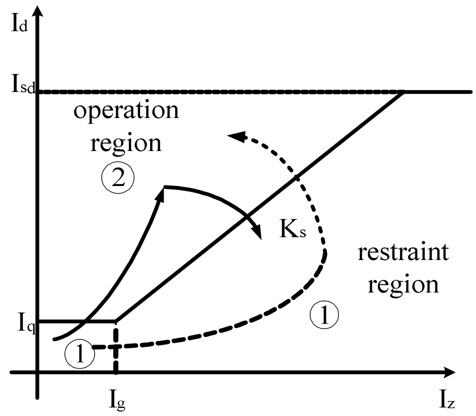



Figure 3. primary current and secondary current of CT transient saturation

Early in internal fault, as CT is in the linear region and the imbalance current can be reflected correctly, the differential current is very large.The increase in self-adaptive filter data window will increase the sampling accuracy, as from window 1 to window 2 in Figure 4(a). But when the CT saturation occurs, differential current gradually decreases. Increasing the data window is not appropriate at this time to the right action of protection, as the increase in the data window contains both CT linear and CT saturation region of current data, as is shown in the data window 3 of Figure 4(a), resulting action point entering the restraint region of protection plane, differential protection of its trajectory is shown in Figure 4(b) by the solid arrow. Early in external fault, CT is in the linear region, and the differential current is very small. With the CT saturation occurs, differential current decreases, but because of the transient characteristics of CT saturation, CT within a cycle will gradually quit saturation, and works in the iron core of the linear region, and differential current decreases. Similarly, when inrush current emerges in the circumstance where none inrush current locking linkage was applied, transformer iron core works in the saturation region where part of pulse of inrush current generally and works in the linear region where part of discontinuous zone of inrush current. But with the dynamic increase in sampling points, the data window will also cover the current of transformer iron core saturation region and linear region, the data of CT is insensitive to detected when CT quits saturation, causing the calculated differential current being large, and protection is prone to malfunction. Differential protection of its trajectory of external fault and inrush current is shown in Figure 4(b) by the dashed arrow.



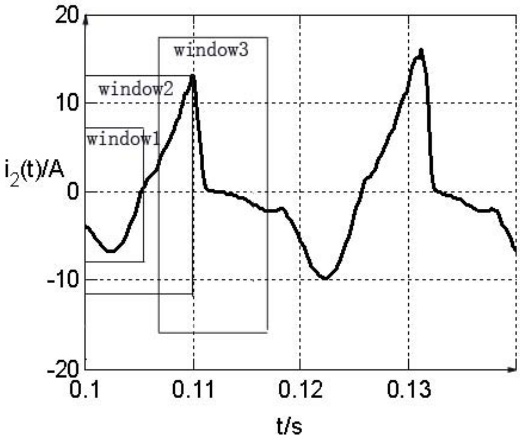

(a) non-adaptive data window diagram



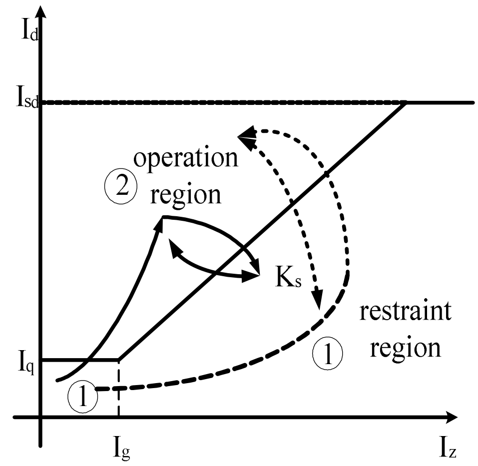

(b) non-adaptive data window action platform

Figure 4. non-adaptive data window diagram and action platform

If the data window dynamically reduced when CT saturation and inrush current occurs, making the length of the data window match the current width when CT is out of transient saturation and returns linear region Figure 5(a),the action points will return to action region again and shuttle between action region and restraint region after CT saturation when internal fault occurs, trajectory shown in Figure 5(b). At this time, based on the two condition of the initial action point being in the action region and the action point keeping shuttling, the protection can rapidly detect an internal fault occurred. When external fault and inrush current occurred, action point will return to restraint region form action region again and shuttle between these two regions. When the length of data window is shorter, inflection current and ratio restraint Coefficient of differential current should be adjusted and coupled with a certain of number of outlet judgment, and the safety of protection will improve greatly.




(a) adaptive data window diagram




(b) adaptive data window action platform

Figure5. adaptive data window diagram and action platform

At this time, recursive least square method can be used to remove the forward data of data window, making the length of the data window match with the current width when CT is out of transient saturation and returns linear region. According to equation (18),it could be inferred that

$$Y(n-k)=Y(n)+K(n-k)\left[X(n-k)-h(n-k)Y(n)\right] \quad (19)$$

where

$$K(n-k)=P(n)h^T(n-k)\left[h(n-k)P(n)h^T(n-k)+1\right]^{-1}$$

, $n$ stands for the data window length on CT's saturation, and $k$ stands for the data window length which should be reduced.

$Y(n-k)$ stands for estimated parameters after dynamically adjusting of the data window length. After removing the forward data, equation (18) and (19) can be used to calculate the following estimates with moving of data window.

## 4.2 Analysis of data window transient

When CT saturation and inrush current occurs, reducing the data window dynamically, and making the length of the data window match with the current width when CT quits transient saturation and return to the linear region can improve the protection performance. Because fault current contains a large number of harmonic components when CT saturation and inrush current occurs, equation (20) can be used to estimate the sampling point of current according to the length of data window at that time, make

$$\tilde{X}=AY=A(A^TA)^{-1}A^TX \quad (20)$$

$X$ is filter matrix of data window length at that time, and $\tilde{X}$ is the estimated result of least squares algorithm. If the data window contains both CT linear and CT saturation region of current data, deviation between $\tilde{X}$ and $X$ would be significant, If the data window contains only CT linear and CT saturation region of current data, $\tilde{X}$ and $X$ would be very similar. So make

$$r=\tilde{X}-X=\left[AA^{+}-I\right]X \quad (21)$$

then

$$r=MX \quad (22)$$

suppose

$$t=\sum_{k=1}^{N}|r_k| \quad (23)$$

$t$ stand for error indicator, when the length of data window is larger than width of linear region, $t$ will be relatively large, length of data window can be reduced dynamically at this time. When the length of data window matches the width of linear region, the value of $t$ will reduce to zero.

Taking it into consideration that when the length of the data window reduce to the linear region width, the number of sampling points will reduce, making the error of the filter increase, so when the data window is crossing saturated region, secondary current can avoid the interference of high harmonics and reflect the primary current more accurately because of the good resistance of recursive least squares algorithm. The differential current of protection will become smaller and the risk of malfunction will be further reduced.

# V. TEST

After analyzing the action properties with simulation data, the model with the d/y-11 transformer is adopted to emulate internal high-voltage side ground fault, external fault when CT saturated and transformer inrush current status to test the algorithm put forward in this paper, as is shown in Figure 6:

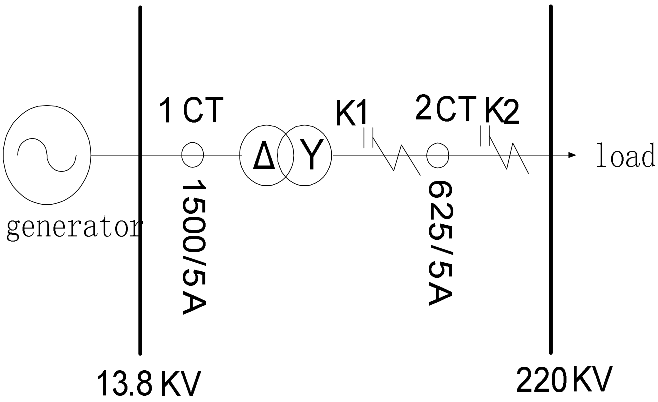
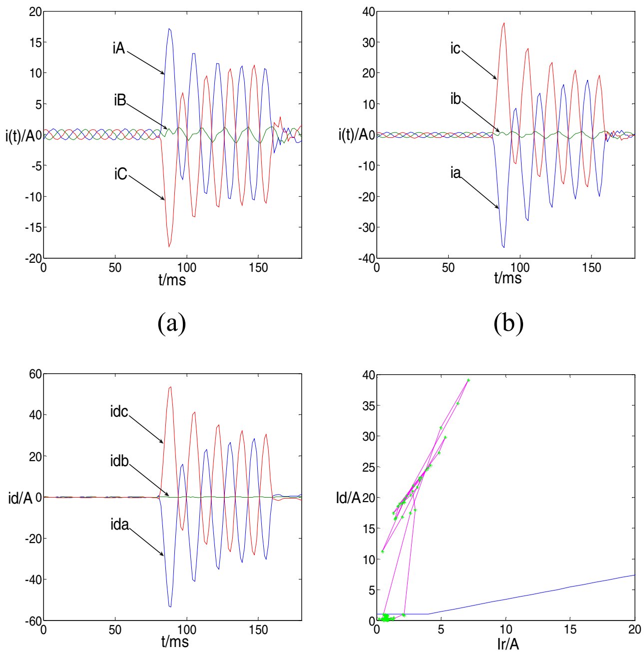



Figure 6. model and parameters of the transformer

The transformer and CT converted current is brought into question, and quarter of the cycle is taken as the initial data of phaselet algorithm, then the number of sampling points is adjusted according to the occurrence of faults.

As is shown in the figure, phase A ground short circuit fault occurred on K1 in the high-voltage side. The primary current and secondary current are also shown in Figure 7, Figure 7(a)/(b) are respectively the momentary current of the low/high voltage side of the transformer, Figure 7(c) shows all the differential currents, Figure 7 (d),(e) and (f) are respectively the action platform of the differential protection algorithm of phase A,B and C. It can be seen that the increase of sampling points can amend the parameter estimation of the initial status and thus significantly improve the calculation accuracy so that protection can operate promptly.



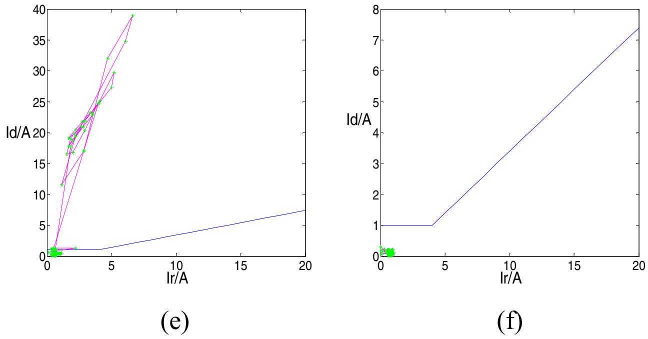
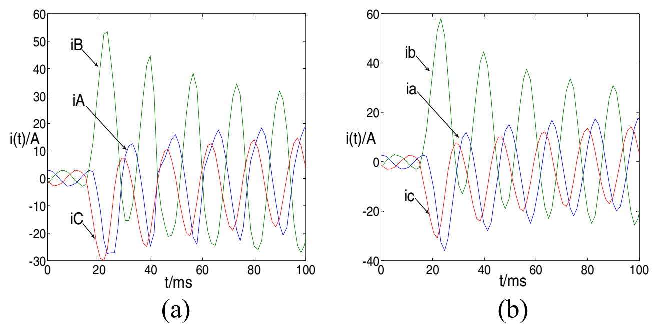
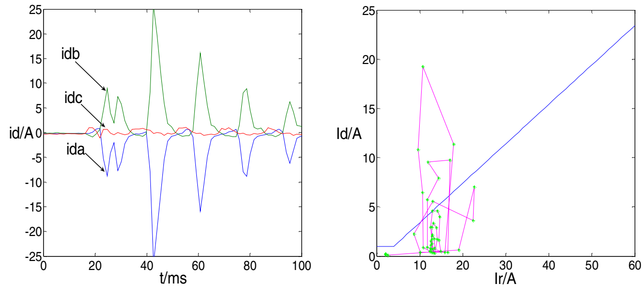



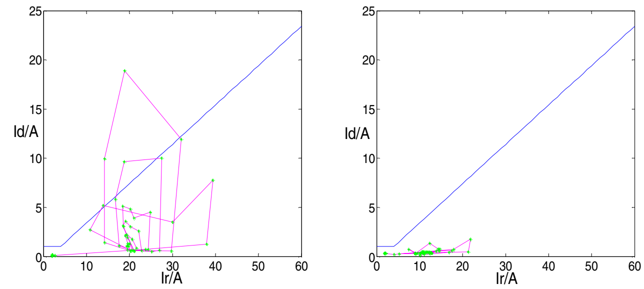

Figure 7. Currents and action platforms on internal faults

The current of low/high voltage side when the two-phase ground fault occurs on K2 of the external high voltage side are shown in Figure 8(a)/(b). It can be inferred that CT is saturated near 20ms and the wave form distorted. Figure 8(c) shows the differential current, it can be seen that protection tends to malfunction because of the continuous vibration of the differential current after 20 ms due to the temporary saturation property of CT. At this time the length of the data window should be shortened according to error indicator in order to match the length of data window with the linear region so as to resist to transient feature of CT saturation. However on the other hand, the accuracy of calculation will drop because of the shortening of the data window, so continuous judgment of outlet number should be adopted in order to improve the reliability of protection. Figure 8(d),(e) and (f) are respectively protection action platform of A,B and C phase when fault occurs. It can be seen that in the beginning of fault, the action point was in the restraint region, then with the occurrence of CT's saturation the action point moved towards the action region and at last with the adjustment of the number of sampling point and the anti-CT saturation process of the algorithm, the action point shuttled between the action region and the restraint region. At this time continuous judgment of outlet number can be taken to action points. As is shown in Figure 8(d), the number of action points towards to the action region won't exceed 5 continuously because of the match of the data window, so 5 points can be used as a criterion of outlet judgment.





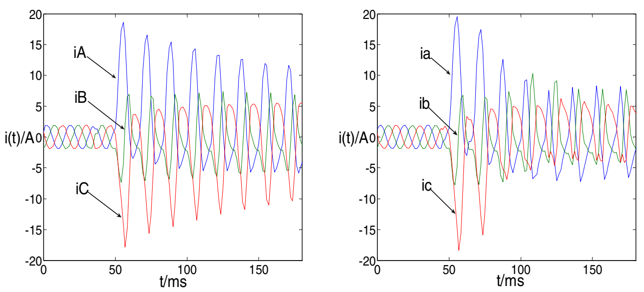

(c) (d)



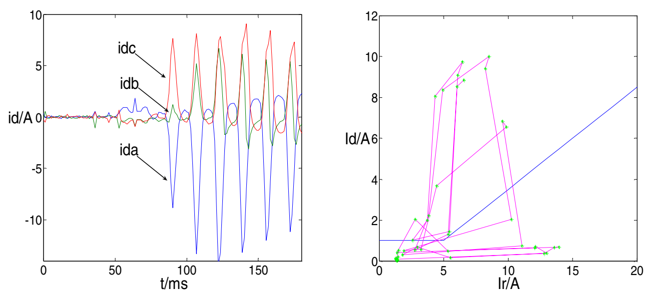

(e) (f)

Figure 8. The currents and action platforms on the external CT saturation

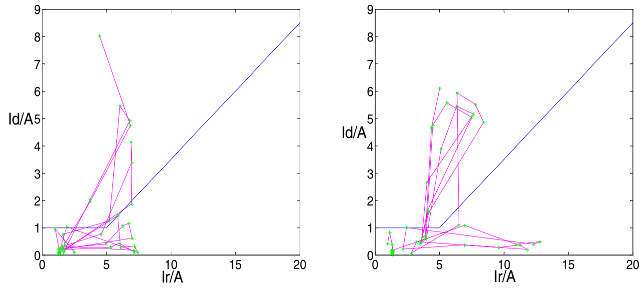
When cross inrush current occurs in a transformer, the low/high voltage side current are shown in Figure 9(a)/(b). At 50 ms, cross inrush current occurred in the transformer, and the wave form distorted. Figure 9(c) shows the differential current, and Figure 9(d),(e) and (f) are the platform of differential protection action of phase A,B and C. CT is transient saturated when the cross inrush current occurs, whose judgment is similar with the external CT.




(a) (b)




(c) (d)




(e) (f)

Figure 9. The currents and action platforms on the transformer inrush current

# VI. Conclusion

CT saturation caused by DC component, external fault and transformer's inrush current will affect the restraint property of the protection significantly, whereas dynamic data window phaselet algorithm based on principle of least squares and certain number of outlet judgment will improve the stability of the protection greatly. It can be seen from the experiment above that under the condition of the dynamic change of the data window, recursive least square algorithm further boosted the protection, the automatic adjustment of data window according to the error indicator would satisfyingly resist the imbalance current brought about by the CT saturation, and the error brought by the short window algorithm can be repaired by the number of outlet judgment. This algorithm is very fit for the stable running of high voltage and extra-high voltage transformers.

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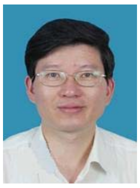

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